

# Image Formation

## Discussion #2

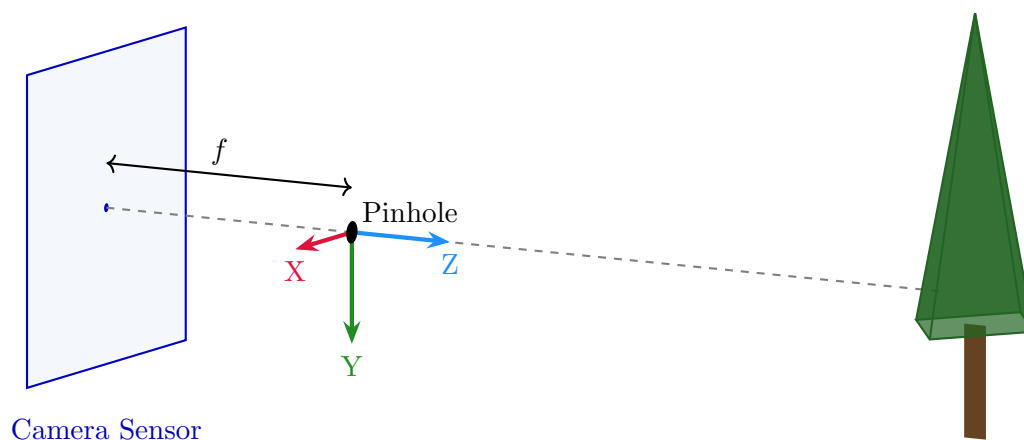
Written by:  
Brent\*, Konpat\*

## Topics

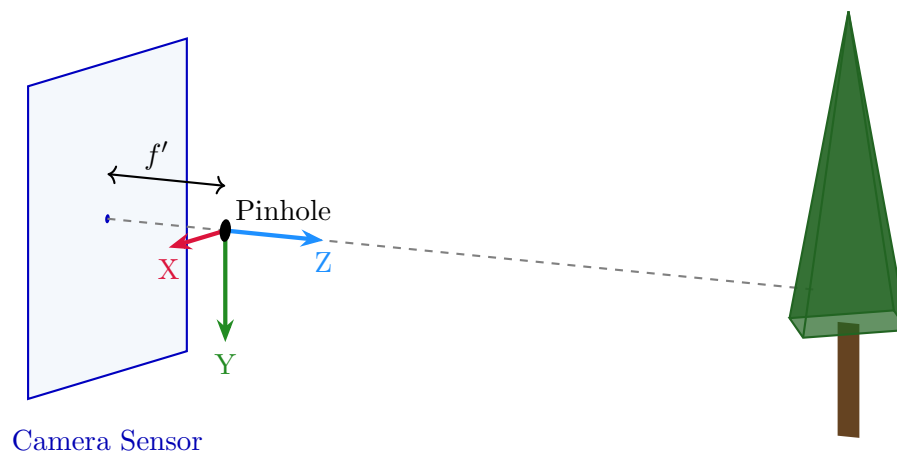
This section covers the pinhole camera model and RGB color.

## 1 Warmup

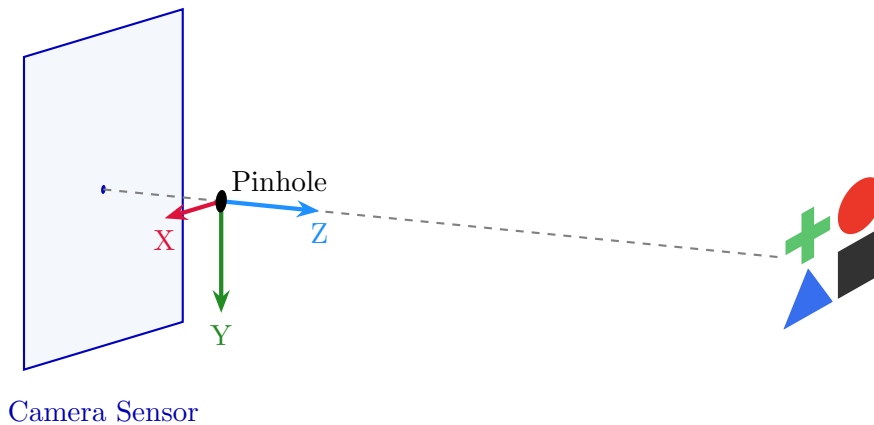
**Problem 1.1:** Chung Min points a pinhole camera with focal length  $f$  directly at a beautiful ponderosa pine tree. How is this object projected onto the image? Draw on the sensor below.



**Problem 1.2:** Unhappy with the original composition, Chung Min shortens the focal length of the camera. How does the image change? Draw on the sensor below.



**Problem 1.3:** Let's try on a harder object. Draw the image as it appears on the sensor below.



**Problem 1.4:** Assume the object has a height of  $H$ , and is located distance  $d$  away from the camera. Given focal length  $f$ , how tall is the projected object on the image plane?

## 2 Dolly Zoom

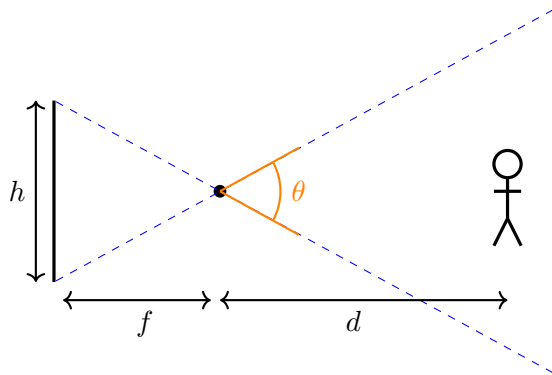
In Project 0, we saw the dolly zoom effect, where multiple camera parameters are adjusted to keep a subject the same size in the image.

**Problem 2.1: *Project 0 recap.*** Between Problem 1.1 and 1.2, Chung Min decided to shorten the camera's focal length. In words, what else needs to be done to achieve a dolly zoom effect?

**Problem 2.2: *How to zoom?*** A camera has initial focal length  $f$ , and is placed a distance  $d$  from a subject. If the camera is moved to distance  $d'$ , what focal length  $f'$  would maintain the subject's size in the image?

**Problem 2.3: *Zoom vs crop.*** Can you achieve the dolly zoom effect without changing the physical focal length of the camera system? Assume  $d' > d$ .

**Problem 2.4: *Angles*.** Real-world computer vision systems often have to grapple with multiple conventions for camera geometry. One that we saw in lecture was field-of-view, which can be expressed in radians as  $\theta$ :



Given initial field-of-view  $\theta$ , initial subject distance  $d$ , updated subject distance  $d'$ , how can we compute updated FOV  $\theta'$  to achieve the dolly zoom effect?

*Hint: you can start from the answer to 2.2, but your final answer should not depend on  $f$  or  $h$ .*

### 3 Color and RGB

A digital color image stores three numbers per pixel:  $(R, G, B)$ . Each channel, viewed on its own, is just a grayscale image that measures how much of that primary is present.

**Problem 3.1: *Channel detective.*** Chung Min photographs a scene containing a red stop sign, green grass, and white clouds. If we display each channel as its own grayscale image, how bright does each object appear in the  $R$ ,  $G$ , and  $B$  channels? Fill in the table with *bright* or *dark*.

|               | $R$ channel | $G$ channel | $B$ channel |
|---------------|-------------|-------------|-------------|
| Red stop sign |             |             |             |
| Green grass   |             |             |             |
| White clouds  |             |             |             |

**Problem 3.2: *Fair and unfair averages.*** A common way to convert a color image to grayscale is the weighted sum

$$Y = 0.299 R + 0.587 G + 0.114 B.$$

Why weight the channels unequally, instead of using the simple average  $(R + G + B)/3$ ?